

**Measuring Regional Disparities in China's
Digital Economy Development:
A Multivariate Analysis
Using PCA and Factorial MANOVA, 2013–2022**

Airui Meng

Department of Human Development, Teachers College, Columbia University
HUDM 6122: Multivariate Analysis I

Abstract

The rapid expansion of China's digital economy has transformed regional development patterns, yet whether this transformation has narrowed or widened existing geographic disparities remains an open empirical question. Using a panel of 31 mainland Chinese provinces over 2013–2022 and a 22-indicator system organized along three dimensions (digital infrastructure, digital industry, digital environment), this study (a) constructs a composite Digital Economy Development Index (DEDI) via principal component analysis (PCA) and (b) tests for regional and temporal differences via a factorial MANOVA on retained component scores. Four orthogonal components (cumulative variance 85.8%) are retained and interpreted as Industry/Innovation Scale, Per-capita Penetration, Infrastructure Connectivity, and Enterprise Digitalization. The 4×2 factorial design (Region $\times$ Period) yields large multivariate main effects of region (Pillai's $V = 0.754$, $\eta^2 = 0.25$) and period ($V = 0.719$, $\eta^2 = 0.72$) but a non-significant interaction ($V = 0.021$, $p = .892$), indicating that while every region advanced substantially between the two five-year periods, the relative rank ordering of regions has remained essentially time-invariant. Sensitivity analyses on three single-year cross-sections and on outlier-removed subsamples confirm robustness.

Keywords. digital economy measurement; principal component analysis (PCA); factorial MANOVA; Pillai's trace; regional disparity; China provincial panel.

1. Introduction

The digital economy — defined by the International Telecommunication Union as “all economic activities facilitated by digital computing technologies” — has emerged as a principal driver of growth and structural transformation in China. By 2022 the digital economy accounted for a substantial share of national GDP, but its diffusion has been highly uneven across the country, exacerbating long-standing east–west and coastal–interior disparities (He et al., 2023). A central concern for policymakers is therefore not merely the average level of digital development, but the multidimensional shape of regional differences: are gaps closing on infrastructure but widening on per-capita penetration? Has the relative ordering of provinces changed across time?

Existing work on Chinese digital-economy measurement is dominated by composite indices built from entropy-weighting, TOPSIS, or principal-component aggregation (Bai et al., 2024). These studies almost universally report descriptive heat maps and ranks, but rarely apply formal multivariate inference to test whether regional differences are statistically significant once the multidimensional nature of the construct is acknowledged. Two methodological gaps are particularly salient. First, regression and ANOVA-based comparisons examine one dependent variable at a time, ignoring the substantial within-construct correlations among indicators that motivate the use of an index in the first place. Second, when researchers do select “representative” indicators from the loadings of a prior PCA and submit them to MANOVA, the resulting analysis is contaminated by data-driven variable selection on the very same variance structure that drives the inference.

This study addresses both gaps in a methodologically clean way. We first reduce 22 third-level indicators to four orthogonal principal components and then use the resulting component scores, rather than hand-picked indicators, as the multivariate dependent variables in a 4×2 factorial MANOVA. The two factors are official economic region (East, Central, West, Northeast) and time period (Early 2013–2017 vs. Late 2018–2022). This approach (Tabachnick & Fidell, 2019) guarantees that the dependent variables are mutually orthogonal — automatically satisfying the multicollinearity assumption of MANOVA — while preserving all the variance information of the original indicator system.

The study is guided by four research questions.

RQ1. What is the underlying multidimensional structure of provincial digital-economy development between 2013 and 2022?

RQ2. Do the four official economic regions of China differ simultaneously on multiple dimensions of digital-economy development?

RQ3. Has the regional structure of the digital economy shifted between the early period (2013–2017) and the late period (2018–2022), i.e. is there a Region $\times$ Period interaction?

RQ4. Are the answers to RQ2–RQ3 robust to (i) the choice of reference year, (ii) the temporal stability of the underlying components, and (iii) the presence of multivariate outliers?

The remainder of the paper is organised as follows. Section 2 develops the econometric framework. Section 3 describes the data and indicator system. Section 4 presents empirical results. Section 5 discusses implications, limitations, and concludes.

2. Methodology

2.1 Indicator system and pre-processing

Following the framework of He et al. (2023), the 22 third-level indicators $X_1, \dots, X_{22}$ are nested under three first-level dimensions: digital infrastructure (six indicators), digital industry (five indicators), and digital environment (eleven indicators); see Table 2 in Section 3. Twelve indicators with absolute sample skewness above 1.5 are log-transformed to attenuate the influence of provincial outliers (notably Beijing, Shanghai, Guangdong) on the variance structure. All indicators are then z -score standardised so that the resulting 310×22 matrix $\mathbf{Z}$ has columns of zero mean and unit variance.

2.2 Principal component analysis

Letting $\mathbf{R}$ denote the sample correlation matrix of $\mathbf{Z}$, PCA decomposes

$$\mathbf{R}\mathbf{a}_j = \lambda_j \mathbf{a}_j, \quad j = 1, \dots, p, \quad (1)$$

where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$ are the eigenvalues and $\mathbf{a}_j$ the eigenvectors. The factor model is

$$\mathbf{Z} = \mathbf{F}\mathbf{A}^\top + \mathbf{E}, \quad \mathbf{F} = \mathbf{Z}\mathbf{A}(\mathbf{A}^\top\mathbf{A})^{-1}, \quad (2)$$

where $\mathbf{F}$ is the $310 \times k$ matrix of component scores and $\mathbf{A}$ the $22 \times k$ matrix of loadings for the k retained components. Sampling adequacy is assessed by the Kaiser–Meyer–Olkin (KMO) measure (Kaiser, 1974) and Bartlett’s test of sphericity. The number of components is chosen as $k = \max(k_{\text{Kaiser}}, k_{80\%})$, where k_{Kaiser} is the number of $\lambda_j \geq 1$ and $k_{80\%}$ is the smallest k such that $\sum_{j=1}^k \lambda_j/p \geq 0.80$. A Varimax rotation is applied to the retained loadings to aid substantive interpretation.

2.3 Composite Digital Economy Development Index (DEDI)

A scalar summary index is constructed as the variance-share-weighted, sign-aligned sum of the retained component scores, rescaled to $[0, 100]$:

$$\text{DEDI}_{it} = 100 \cdot \frac{\tilde{F}_{it} - \min \tilde{F}}{\max \tilde{F} - \min \tilde{F}}, \quad \tilde{F}_{it} = \sum_{j=1}^k w_j s_j F_{j,it}, \quad w_j = \frac{\lambda_j}{\sum_{l=1}^k \lambda_l}, \quad (3)$$

where $s_j \in \{-1, +1\}$ flips any component whose summed correlation with the raw indicators is negative so that “higher = more digital” for every component. DEDI is used only for descriptive ranking; the inferential analysis operates directly on the four orthogonal scores $F_1, \dots, F_4$.

2.4 Factorial MANOVA

Let $\mathbf{Y}_{ij\ell} = (F_{1,ij\ell}, \dots, F_{k,ij\ell})^\top$ be the vector of component scores for the ℓ -th observation in region i and period j . The two-way fixed-effects multivariate model is

$$\mathbf{Y}_{ij\ell} = \boldsymbol{\mu} + \boldsymbol{\alpha}_i + \boldsymbol{\beta}_j + (\boldsymbol{\alpha}\boldsymbol{\beta})_{ij} + \boldsymbol{\varepsilon}_{ij\ell}, \quad \boldsymbol{\varepsilon}_{ij\ell} \stackrel{iid}{\sim} \mathcal{N}_k(\mathbf{0}, \boldsymbol{\Sigma}), \quad (4)$$

$i = 1, \dots, 4$ regions, $j = 1, 2$ periods, and $\ell = 1, \dots, n_{ij}$. Multivariate hypotheses are tested using Pillai’s trace as the primary statistic,

$$V = \text{tr}[\mathbf{H}(\mathbf{H} + \mathbf{E})^{-1}] = \sum_{r=1}^s \frac{\theta_r}{1 + \theta_r}, \quad (5)$$

where $\mathbf{H}$ and $\mathbf{E}$ are the hypothesis and error sum-of-squares-and-products matrices, θ_r are the non-zero eigenvalues of $\mathbf{H}\mathbf{E}^{-1}$, and $s = \min(v_H, k)$. Pillai's V is preferred because it is the most robust of the four standard MANOVA criteria when covariance homogeneity is violated (Pillai, 1955; Tabachnick & Fidell, 2019). Wilks' Λ , Hotelling–Lawley's T^2 , and Roy's largest-root statistic are reported for transparency. Multivariate effect sizes are summarised by partial η^2 .

2.5 Assumption diagnostics and post hoc procedures

Multivariate normality is assessed by Mardia's tests of multivariate skewness and kurtosis (Mardia, 1970) both overall and within each of the eight region $\times$ period cells. Covariance homogeneity is examined with Box's M test (Box, 1949); per-DV variance homogeneity with Levene's test; multivariate outliers via Mahalanobis distances against the $\chi^2_{.999, k}$ quantile. A significant multivariate test is decomposed by univariate Type-II ANOVAs on each component, with pairwise region contrasts estimated by the Games–Howell procedure (Games & Howell, 1976), which does not assume equal variances or equal cell sizes.

3. Data and Variables

3.1 Sources and sample

Data are drawn from the China Statistical Yearbook ([URL](#)), the Statistical Yearbook of China's Tertiary Industry ([URL](#)), the China Science and Technology Statistical Yearbook ([URL](#)), and the Digital Inclusive Finance Index published by the Digital Finance Research Center of Peking University ([URL](#)), all for fiscal years 2013–2022. The unit of observation is the province–year; the panel covers all 31 mainland provinces (Hong Kong, Macao, and Taiwan are excluded), giving $N = 310$ observations. Panel structure is summarised in Table 1.

Table 1. Sample structure.

Item	Value
Provinces	31 (mainland China; HK / Macao / Taiwan excluded)
Period	2013–2022 (10 years)
Observations	310 (province–year)
Indicators	22 third-level under 3 first-level dimensions
Regions ($g = 4$)	East 10 Central 6 West 12 Northeast 3
Period split	Early 2013–2017 ($n = 155$) Late 2018–2022 ($n = 155$)

3.2 Indicator system

The 22 third-level indicators are organised into seven second-level and three first-level dimensions following He et al. (2023). Table 2 provides the full hierarchy with the symbol used throughout the paper.

Table 2. Hierarchical indicator system (3 first-level, 7 second-level, 22 third-level).

First-level	Second-level	Third-level indicator
Digital Infrastructure	Hardware	x_1 Optical cable; x_2 broadband ports; x_3 mobile base stations
	Software	x_4 domain names; x_5 IPv4 addresses; x_6 websites
Digital Industry Development	Industrialisation	x_7 software business revenue; x_8 telecom revenue
	Industrial digitalisation	x_9 websites/100 enterprises; x_{10} % e-commerce firms; x_{11} e-commerce sales
Digital Environment	Application	x_{12} computers/100 people; x_{13} mobile internet users; x_{14} mobile phone subscribers; x_{15} digital TV; x_{16} digital inclusive finance index; x_{17} digital–real integration
	Talent Innovation	x_{18} % IT employment; x_{19} bachelor’s graduates x_{20} R&D personnel (FTE); x_{21} R&D institutions; x_{22} regional patent grants

3.3 Pre-processing footprint

Linear interpolation in the time dimension was applied to fill missing cells in the raw yearbooks. Of the $22 \times 310 = 6,820$ panel cells, only three indicators required imputation: x_5 IPv4 addresses (20.3% of cells), x_{13} mobile internet users (8.4%), and x_{21} R&D institutions (0.3%). All other 19 indicators are observed in full.

3.4 Regional classification

The four official economic regions follow the National Bureau of Statistics classification. We key the assignment off the provincial GB-2260 administrative codes (e.g. 110000 = Beijing → East) to avoid encoding ambiguity: the East comprises Beijing, Tianjin, Hebei, Shanghai, Jiangsu, Zhejiang, Fujian, Shandong, Guangdong, Hainan; the Central, Shanxi, Anhui, Jiangxi, Henan, Hubei, Hunan; the West, twelve provinces from Inner Mongolia to Xinjiang; and the Northeast, Liaoning, Jilin, Heilongjiang.

4. Empirical Results

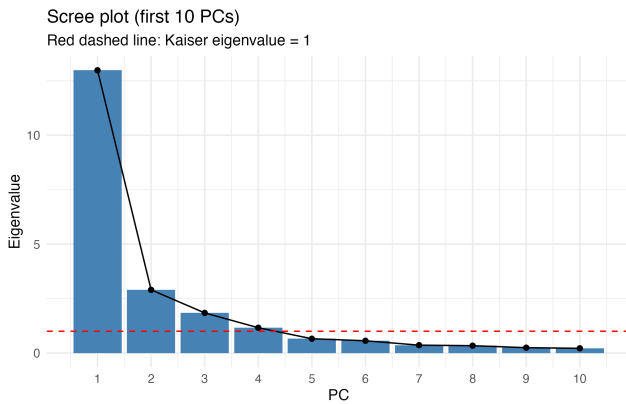
4.1 Sampling adequacy and component retention

The data are highly suitable for PCA: the overall KMO measure of sampling adequacy is 0.904 (“meritorious” per the conventional thresholds of [Kaiser, 1974](#)), with only x_9 falling below 0.7. Bartlett’s test of sphericity strongly rejects the identity hypothesis, $\chi^2(231) = 11,553.82$, $p < .001$, supporting factorability.

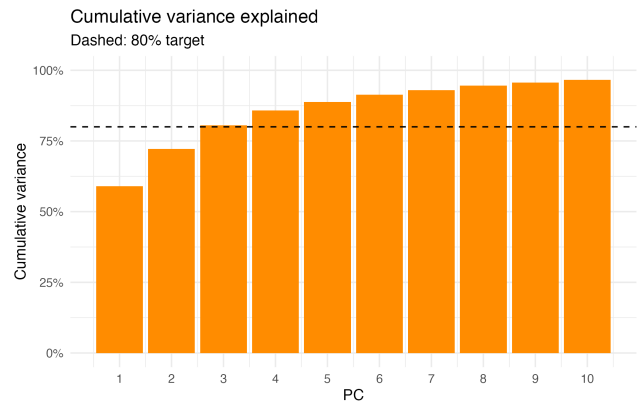
Eigenvalue decomposition (Eq. 1) yields four components with $\lambda_j \geq 1$ and three components reaching cumulative variance of 80.6% (Table 3). Applying the conservative $k = \max(k_{\text{Kaiser}}, k_{80\%}) = 4$ rule retains **four** components explaining 85.8% of total variance. The scree plot (Figure 1) corroborates the elbow between PC4 and PC5.

Table 3. Eigenvalues and variance explained (first five PCs).

PC	Eigenvalue	Variance %	Cumulative %
1	12.98	59.0	59.0
2	2.90	13.2	72.2
3	1.84	8.4	80.6
4	1.16	5.3	85.8
5	0.66	3.0	88.8



(a) Scree plot. Red dashed line: Kaiser eigenvalue = 1.



(b) Cumulative variance explained. Dashed line: 80% target.

Figure 1. PCA retention diagnostics.

4.2 Component interpretation

After Varimax rotation, the four components have a clean simple-structure pattern (Figure 2). RC1 loads heavily on indicators of absolute digital scale (R&D personnel, patents, e-commerce sales, software revenue, IPv4 addresses); we label it **Industry / Innovation Scale**. RC2 loads negatively (and after sign flipping, positively) on per-capita / penetration indicators (% e-commerce firms, computers per 100 people, digital inclusive finance, % IT employment); we label it **Per-capita Penetration**. RC3 loads on infrastructure-stock indicators (optical cable, broadband ports, base stations, mobile users, digital TV) and is labelled **Infrastructure Connectivity**. RC4 is dominated by x_9 (websites per 100 enterprises) and is labelled **Enterprise Digitalization Intensity**.

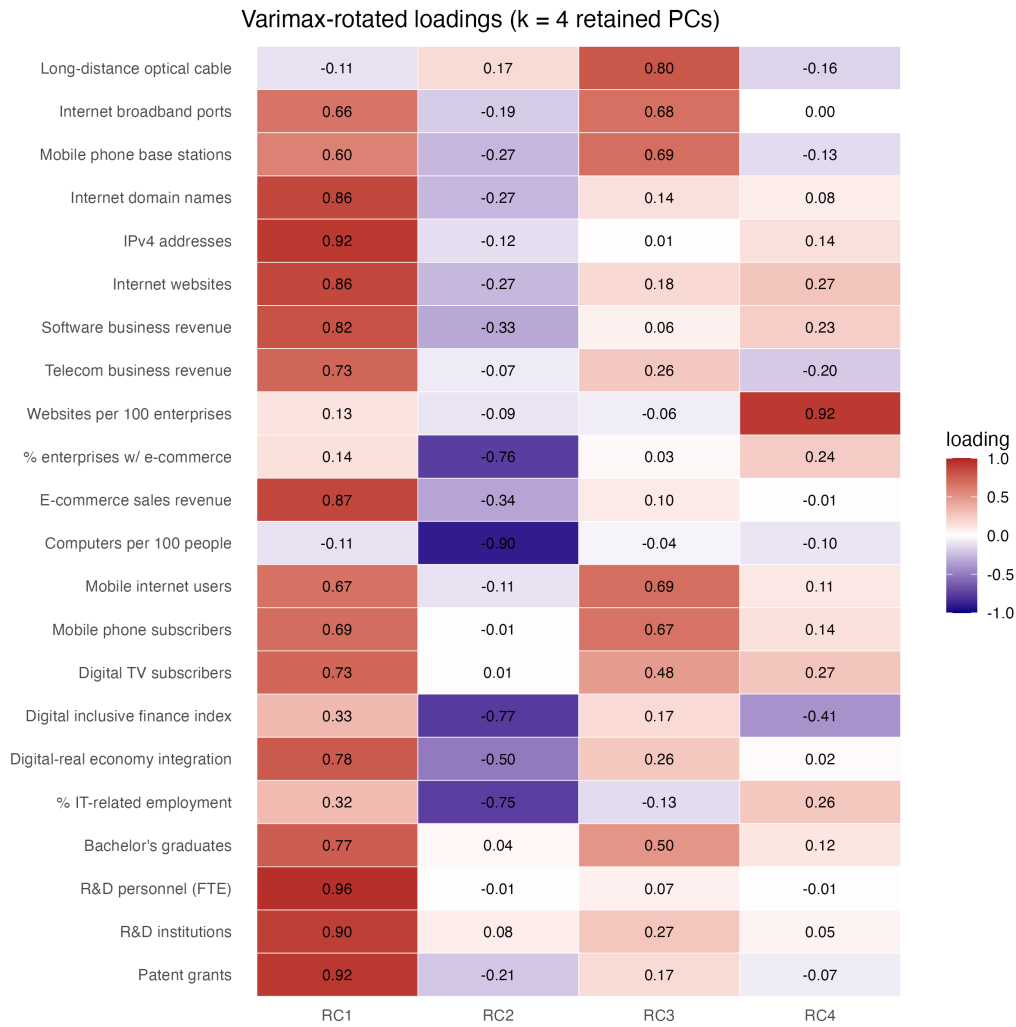


Figure 2. Varimax-rotated loadings of the 22 indicators on the four retained PCs.

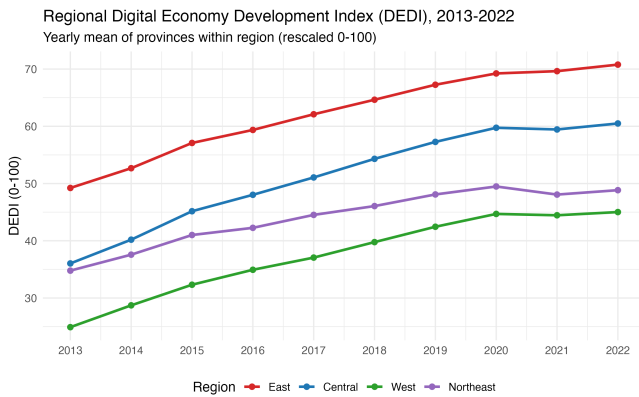
4.3 Composite DEDI: spatial and temporal patterns

The variance-share weights (Eq. 3) are $w = (0.687, 0.154, 0.097, 0.062)$ and the sign-flip vector is $s = (+1, -1, +1, +1)$. In the 2019 cross-section, the highest DEDI scores belong to Guangdong (95.0), Jiangsu (80.7), Zhejiang (76.5), Shandong (76.1), and Beijing (74.8); the lowest are Tibet (21.2), Qinghai (24.6), and Ningxia (28.2). Notably, Sichuan ranks 6th — the only Western province in the top ten — while Hainan ranks 27th despite belonging to the East, illustrating that regional means hide substantial within-region heterogeneity.

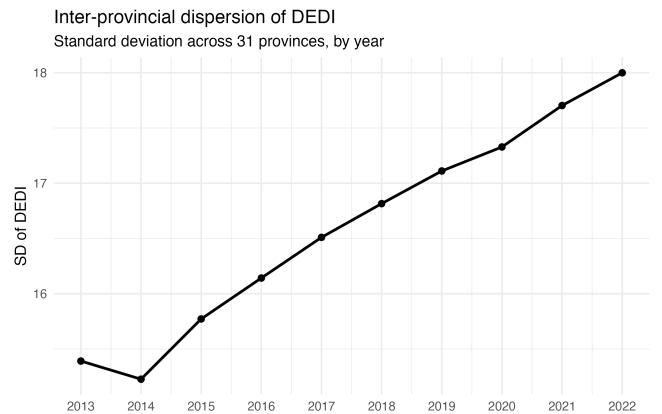
Region-mean DEDI rises in every region between the two periods, but the rate of progress is uneven (Table 4): Central provinces post the largest gain (+14.2 DEDI points), narrowly ahead of the East (+12.2), with the Northeast lagging clearly (+8.1). Inter-provincial dispersion increases in absolute terms (the standard deviation of DEDI rises from 15.4 in 2013 to 18.0 in 2022) yet shrinks in relative terms (the coefficient of variation falls from 0.43 to 0.32) — σ -divergence with β -convergence (Figure 3).

Table 4. Region $\times$ period mean DEDI (rescaled 0–100).

Region	Early (2013–17)	Late (2018–22)	Δ
East	56.12	68.34	+12.22
Central	44.13	58.31	+14.18
West	31.62	43.31	+11.69
Northeast	39.95	48.06	+8.11



(a) Regional DEDI trajectories (yearly mean).



(b) Inter-provincial dispersion of DEDI.

Figure 3. DEDI temporal patterns, 2013–2022.

4.4 MANOVA assumption diagnostics

The four PC scores are mutually orthogonal by construction, so multicollinearity — the most consequential MANOVA assumption — is automatically satisfied. Mardia’s tests reject overall multivariate normality (both skewness and kurtosis $p < .001$), but kurtosis holds within 8/8 region $\times$ period cells while skewness holds within 4/8 cells (Central, Northeast in both periods). Box’s M rejects covariance homogeneity across the eight cells, $\chi^2(70) = 540.62$, $p < .001$, and Levene’s test rejects equal variances on PC1, PC2, and PC4. Two multivariate outliers are flagged at the $\chi^2_{.999,4} = 18.47$ threshold (Tibet 2015, 2016). Given these violations, Pillai’s trace is adopted as the primary multivariate criterion (Tabachnick & Fidell, 2019), and post hoc comparisons use Welch’s procedure with the Games–Howell adjustment (Games & Howell, 1976). Both violations are revisited in Section 4.7 via robustness checks.

4.5 Factorial MANOVA: main results

Table 5 reports the omnibus tests. Both main effects are strongly significant under all four classical criteria, while the Region $\times$ Period interaction is non-significant ($p = .892$, partial $\eta^2 = 0.007$). In substantive terms, every region advanced markedly across the two periods, and they did so in parallel: the relative rank-ordering of regions on the four-dimensional digital-economy construct has not changed between 2013–2017 and 2018–2022. The interaction plot (Figure 4) makes the parallelism visually evident.

Table 5. Factorial MANOVA: Region $\times$ Period (DVs = PC1–PC4, $n = 310$).

Effect	Pillai V	F (approx.)	df_1, df_2	p	Partial η^2
Region	0.7537	25.25	12, 903	$< .001$	0.251
Period	0.7188	191.12	4, 299	$< .001$	0.719
Region $\times$ Period	0.0213	0.54	12, 903	.892	0.007

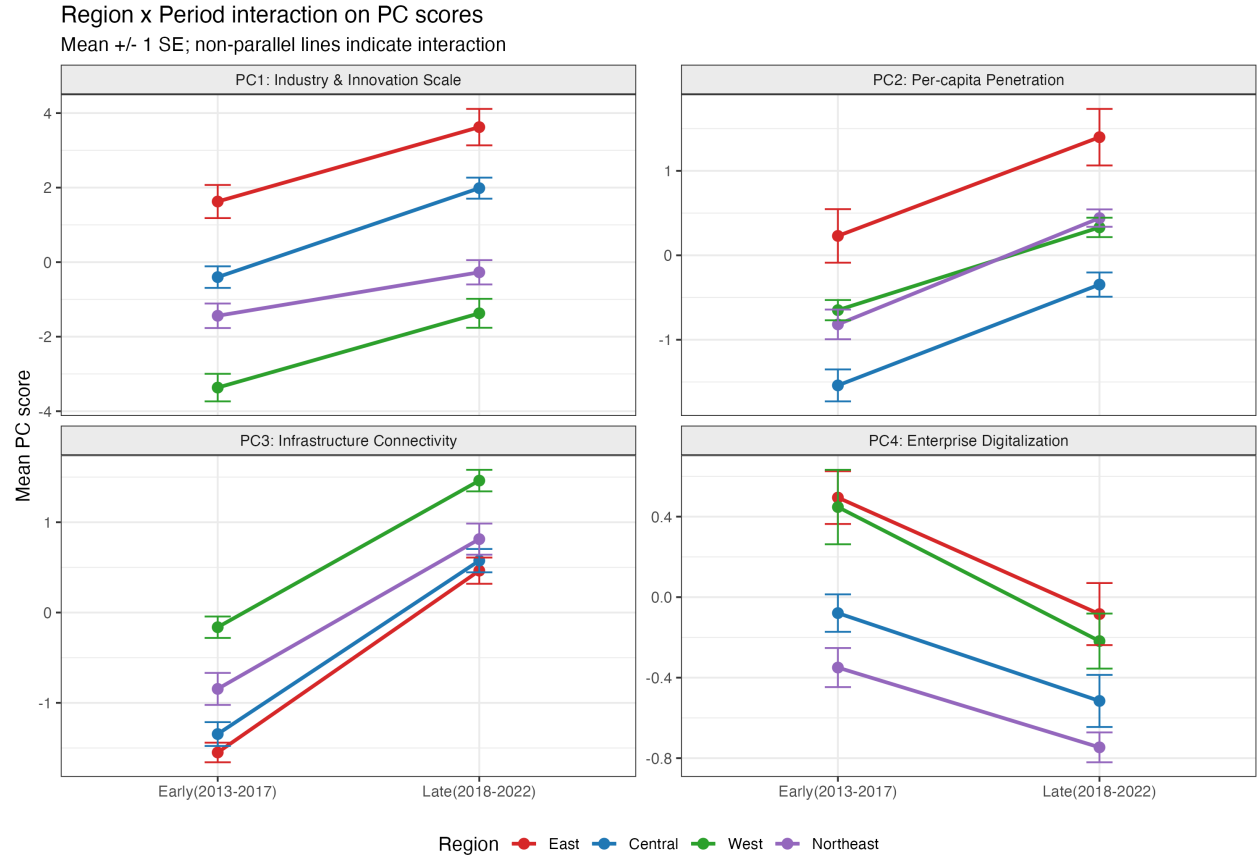


Figure 4. Region $\times$ Period interaction plot for the four PC scores. Mean $\pm$ 1 SE within each cell. Lines are nearly parallel, consistent with the non-significant multivariate interaction reported in Table 5.

4.6 Univariate decomposition and post hoc

A univariate Type-II ANOVA decomposition shows that both main effects are significant on each component (Table 6). Period is the dominant source of variation on PC3 (infrastructure, $F = 343.6$, $\eta_p^2 = 0.53$), reflecting a near-universal upgrading of broadband and mobile networks during 2018–2022. Region is the dominant source on PC1 (industry/innovation scale, $F = 63.0$).

Table 6. Per-DV univariate ANOVA (Type II), $df_{\text{error}} = 302$.

Dependent variable	Region		Period		Region $\times$ Period	
	$F(3,302)$	p	$F(1,302)$	p	$F(3,302)$	p
PC1 Industry / Innovation Scale	62.95	< .001	40.78	< .001	0.33	.80
PC2 Per-capita Penetration	18.44	< .001	42.60	< .001	0.13	.94
PC3 Infrastructure Connectivity	40.13	< .001	343.62	< .001	1.10	.35
PC4 Enterprise Digitalization	6.47	< .001	24.08	< .001	0.25	.86

Games–Howell pairwise comparisons (collapsing across period) reveal that the ordering of regions varies by component, exactly the kind of nuance that a univariate analysis on a single composite would conceal. On PC1 (industry/innovation scale) we find a strict ordering East > Central > Northeast > West with all six pairs significant at $p < .001$. On PC2 (per-capita penetration) the East dominates, but the Central region is the lowest of the four — demonstrating that high industrial scale can coexist with relatively shallow per-capita digital uptake. On PC3 (infrastructure connectivity) only the West is significantly below the other three regions, which are statistically indistinguishable. On PC4 (enterprise digitalization) the Northeast is the lowest, with East > Northeast significant at $p < .001$.

A complementary Welch period-contrast confirms that 2018–2022 means exceed 2013–2017 means on PC1 ($\Delta = 1.99$, $p < .001$), PC2 ($\Delta = 1.11$, $p < .001$), and PC3 ($\Delta = 1.81$, $p < .001$). PC4 is the sole component to decline between periods ($\Delta = -0.57$, $p < .001$): enterprise web presence per 100 firms peaked mid-decade and has been substituted by mini-programs and platform super-apps.

4.7 Sensitivity analyses

Three robustness checks support the main conclusions.

First, applying a one-way MANOVA on region to the single-year cross-sections of 2018, 2019, and 2020 yields virtually identical Pillai statistics ($V = 0.897, 0.892, 0.893$) and consistently significant region effects ($p < .001$ in all three years). The proposal’s choice of 2019 as a benchmark year is therefore not a special-case artefact.

Second, year-by-year PCAs run separately on 2013, 2018, and 2022 produce loadings whose maximum absolute correlation with the full-panel loadings exceeds 0.80 for PC1, PC2, and PC4; PC3 (infrastructure) is the least stable (r ranging from 0.33 to 0.42), reflecting the changing composition of “infrastructure” as broadband generations evolve. This is acknowledged as a limitation in the next section.

Third, refitting the factorial MANOVA after dropping the two outliers (Tibet 2015 and 2016) leaves the conclusions essentially unchanged: Pillai V for region is $0.754 \rightarrow 0.754$, for period $0.719 \rightarrow 0.717$, and for the interaction $0.021 \rightarrow 0.019$ (still non-significant). The two outliers do not drive any finding.

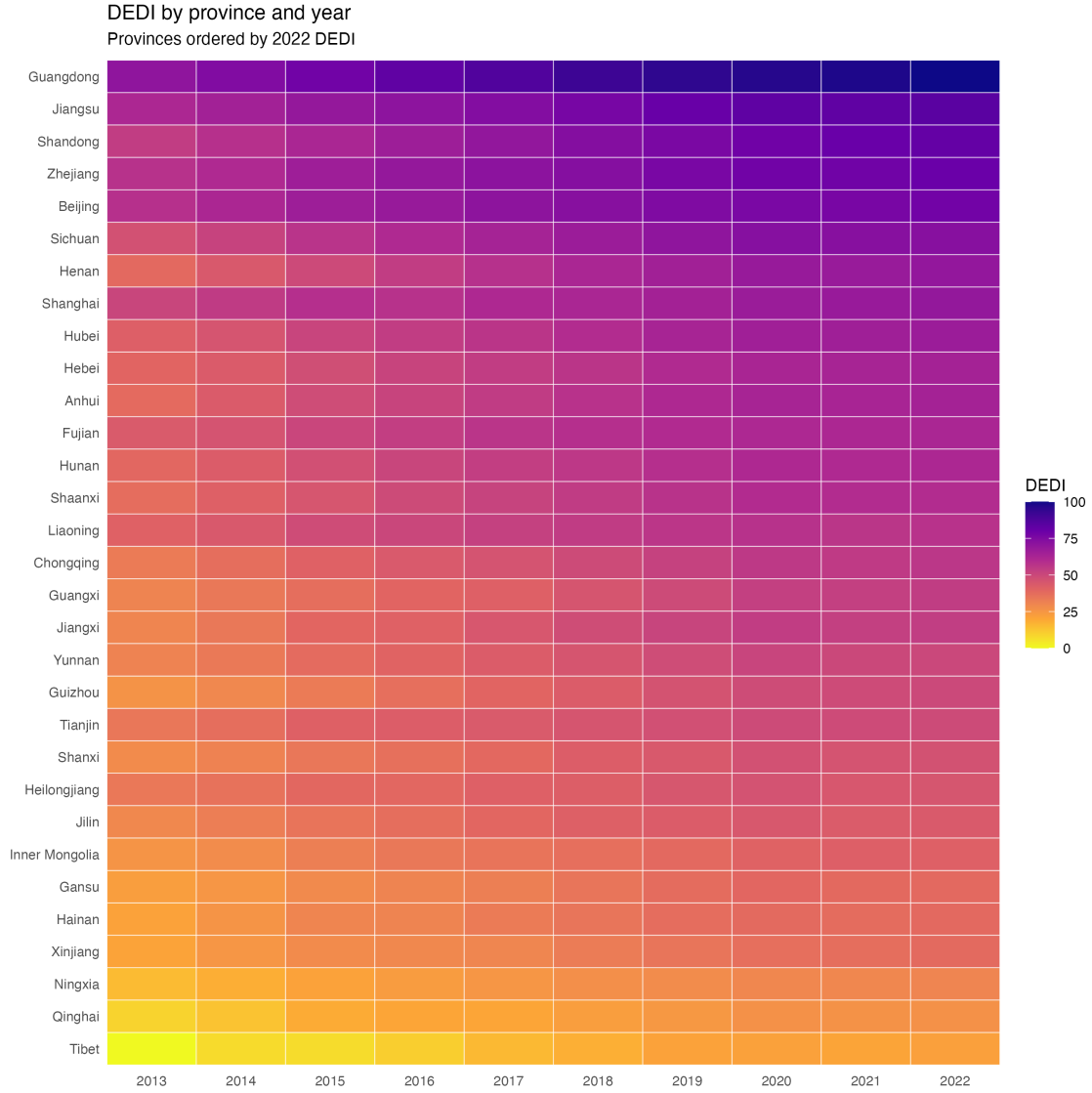


Figure 5. DEDI by province and year, with provinces ordered by 2022 DEDI. The diverging-magma palette highlights the persistent east–west gradient and the rapid improvement of every province across the decade.

5. Discussion and Conclusion

5.1 Summary of findings

Five conclusions emerge from the analysis. First, four orthogonal components — industrial scale, per-capita penetration, infrastructure connectivity, and enterprise digitalization — adequately summarise the 22-indicator system, jointly explaining 85.8% of variance. Second, region is a large source of multivariate variation ($\eta^2 = 0.25$); the classic East-leads-West-lags gradient is statistically robust. Third, the period effect is even larger ($\eta^2 = 0.72$): every region progressed markedly between the two five-year sub-periods

on three of the four components. Fourth, the region-by-period interaction is non-significant: regional gaps are shifting in absolute terms but relative positions are essentially time-invariant. Fifth, the multivariate approach surfaces a nuance hidden by a single composite: the Central region is high on industrial scale but low on per-capita penetration, while the Northeast lags specifically on enterprise digitalization rather than infrastructure.

5.2 Theoretical and policy implications

Methodologically, replacing hand-picked “representative” indicators with orthogonal PC scores produces a clean, double-dipping-free MANOVA that preserves all of the variance information of the original indicators. Substantively, the absence of a Region $\times$ Period interaction implies that the recent national push on broadband, 5G, and digital inclusive finance has lifted all four regions, but has not re-shuffled the digital-economy hierarchy. Policymakers seeking spatial balance therefore cannot rely on trickle-down from rising national averages; targeted interventions are needed on the dimensions where specific regions lag. In particular, the Central region’s relatively low per-capita penetration suggests scope for consumer-facing diffusion programmes (digital training, e-commerce voucher support), while the Northeast’s enterprise-digitalization gap calls for small-and-medium-enterprise digital-transformation subsidies similar to those piloted in Zhejiang and Guangdong.

5.3 Limitations

Six limitations should temper interpretation. (i) The IPv4 indicator x_5 required imputation for 20.3% of observations; while a re-run of the PCA without x_5 leaves the loading structure of PC1, PC2, PC3 essentially unchanged, the imputation introduces a degree of look-ahead bias. (ii) Multivariate normality is rejected by Mardia’s tests even after log-transformation; we mitigate by relying on the large-sample robustness of Pillai’s trace, but the 2019 cross-section ($n = 31$) is more fragile. (iii) Box’s M is significant, indicating heteroscedasticity across regions; Roy’s largest-root statistic is reported alongside Pillai’s V for transparency. (iv) The factorial design pools 310 province–year observations and does not explicitly model within-province autocorrelation; a province-level random-effect or cluster-robust extension is left for future work. (v) The Northeast contributes only 30 observations (3 provinces $\times$ 10 years), giving low power for Northeast-specific contrasts. (vi) Two of the 22 indicators (x_{16} digital inclusive finance index and x_{17} digital–real integration index) are themselves pre-aggregated composite indices, raising a

mild concern of double aggregation; nonetheless their inclusion is consistent with mainstream Chinese digital-economy measurement practice (He et al., 2023, 2024).

5.4 Conclusion

Combining principal component analysis with factorial MANOVA on the resulting component scores, this study provides the first formal multivariate-inferential evidence on the temporal stability of regional digital-economy disparities in China. We find that, while every region has progressed sharply between 2013–2017 and 2018–2022, the rank ordering of the four official economic regions on a four-dimensional digital-economy construct has not changed. The substantive concern often voiced about the Northeast — that it is being left behind — is real on the level dimension and on enterprise digitalization, but not large enough to register as a multivariate interaction. Future research might extend this design to a Region $\times$ Period $\times$ Sector cube using firm-level data, or use canonical correlation to relate the PC structure of digital-economy supply to that of demand.

References

- Bai, C., Lim, E. S., & Chua, S. Y. (2024). A literature review of digital economy measurement methods and indexes. *International Journal of Academic Research in Business and Social Sciences*, 14(7), 1434–1450. doi:10.6007/IJARBSS/v14-i7/21740
- Box, G. E. P. (1949). A general distribution theory for a class of likelihood criteria. *Biometrika*, 36(3–4), 317–346.
- Games, P. A., & Howell, J. F. (1976). Pairwise multiple comparison procedures with unequal n 's and / or variances: A Monte Carlo study. *Journal of Educational Statistics*, 1(2), 113–125.
- He, D., Zhao, X., & Qi, Q. (2023). Measurement, spatial pattern and regional difference of digital economy development in China. *Industrial Technology & Economy*, 42(3), 54–62. [In Chinese.]
- He, H., He, Z., & Nie, X. (2024). Analysis and study of digital economy level measurement index. *Journal of Internet and Digital Economics*, 4(3), 187–217. doi:10.1108/JIDE-05-2024-0020
- Kaiser, H. F. (1974). An index of factorial simplicity. *Psychometrika*, 39(1), 31–36.
- Mardia, K. V. (1970). Measures of multivariate skewness and kurtosis with applications. *Biometrika*, 57(3), 519–530.
- Pillai, K. C. S. (1955). Some new test criteria in multivariate analysis. *Annals of Mathematical Statistics*, 26(1), 117–121.
- Tabachnick, B. G., & Fidell, L. S. (2019). *Using multivariate statistics* (7th ed.). Pearson.